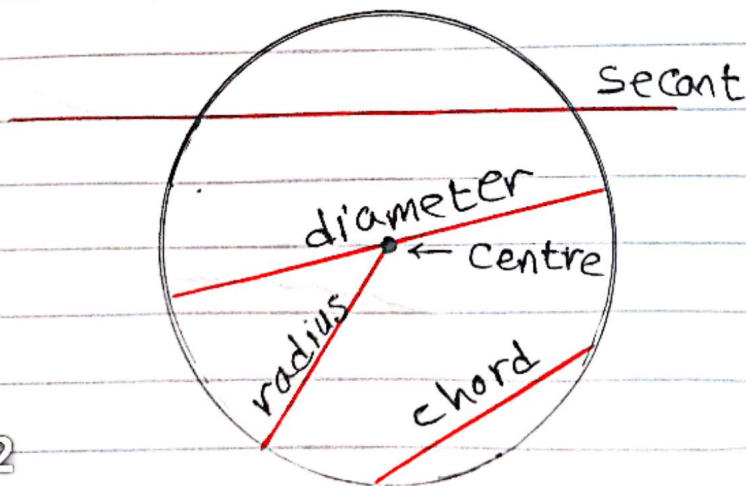


The geometry of a circle.

Def: . A Circle is the set of all points in the Plane that are a fixed distance (the radius) from a fixed point (the center).

- Any interval joining a point on the circle to the center is called a radius.
By the definition of a circle, any two radii have the same length
- Any interval joining two points on the circle is called a Chord
- A chord that passes through the center is called a diameter
$$\text{diameter} = 2(\text{radius})$$
$$d = 2r$$
- A Line that cuts a circle in two distinct points is called a Secant.



Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٢ م

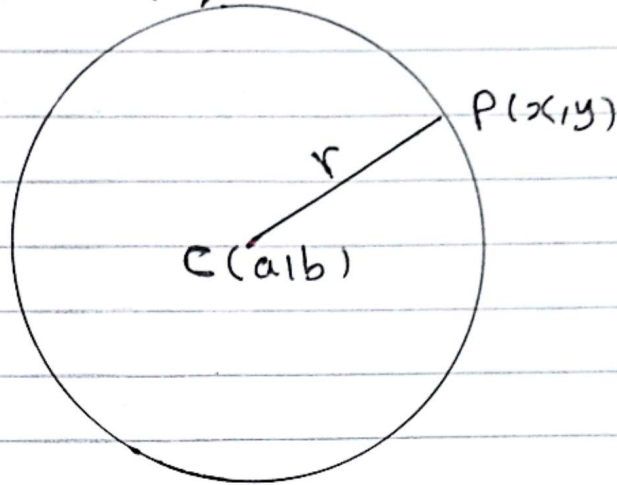
-1-

No:

Date:

The equation of the circle

Def: The equation of the circle of radius r centered at the point $C(a, b)$ is



$$\underline{(x-a)^2 + (y-b)^2 = r^2}$$

Expanding the brackets gives

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 - r^2 = 0$$

$$\text{i.e. } x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) = 0$$

$$\text{put } -a = f, -b = g, a^2 + b^2 - r^2 = c.$$

Thus the equation becomes

$$\underline{x^2 + y^2 + 2fx + 2gy + c = 0}$$

Galaxy M12 This is the general equation of a circle.

٢٩ يوليو ٢٠٢٥ ١٢:٢٢ م

We can recognise it, because it is quadratic in both x and y and it has two additional properties. First, there is no term in xy . And secondly, the coefficient of x^2 is the same as the coefficient of $y^2 = 1$.

The center of the circle $(-f, -g)$

$$\text{radius } r = \sqrt{f^2 + g^2 - c}$$

$\therefore$ the general equation of the circle is

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

where center $(-f, -g)$

$$r = \sqrt{f^2 + g^2 - c}$$

If $f^2 + g^2 - c > 0$ the circle is real

If $f^2 + g^2 - c < 0$ // // // imaginary

If $f^2 + g^2 - c = 0 \Rightarrow r = 0$

and the equation represent the point $(-f, -g)$ (center).

Ex 1

write the equation of the circle with its center at a point $(-2, 3)$ and radius 4

Solution

$$(x-a)^2 + (y-b)^2 = r^2 \Rightarrow$$

$$\Rightarrow (x - (-2))^2 + (y - 3)^2 = 16$$

$$\Rightarrow x^2 + y^2 + 4x - 6y - 3 = 0$$

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٢ م

-3-

No:

Date:

Ex. For the circle whose equation is

$$x^2 + y^2 - 3x + 5y - 14 = 0,$$

Find the coordinates of the Center and the value of the radius.

Sol:

General form

$$x^2 + y^2 + 2fx + 2gy + C = 0$$

$$\text{Center } (-f, -g)$$

$$r = \sqrt{f^2 + g^2 - C}$$

⇒

$$2f = -3 \Rightarrow f = -3/2$$

$$2g = 5 \Rightarrow g = 5/2, \quad C = -14$$

$$r = \sqrt{f^2 + g^2 - C} = \sqrt{\frac{9}{4} + \frac{25}{4} + \frac{56}{4}} = \sqrt{\frac{90}{4}} \\ = \frac{3\sqrt{10}}{2}$$

$$\therefore \text{Center} = \left(\frac{3}{2}, -\frac{5}{2} \right)$$

$$r = \frac{3\sqrt{10}}{2}$$

Ex. Determine the value of K so that

$$x^2 + y^2 - 8x + 10y + K = 0$$

is the equation of a circle whose radius is 7

Sol:

$$\text{Galaxy M12} = \sqrt{f^2 + g^2 - K}, \quad f = -4, \quad g = 5, \quad r = 7$$

$$\sqrt{16 + 25 - K} = \sqrt{41 - K} \Rightarrow 49 = 41 - K \Rightarrow K = -8$$

Ex.

Find the equation of the circle whose center is $(5, -2)$ and which passes through point $(-1, 5)$.

Solution

General form

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

$$\text{Center } (5, -2) \Rightarrow -f = 5, -g = -2$$

$$\Rightarrow f = -5, g = 2 \Rightarrow$$

$$x^2 + y^2 - 10x + 4y + c = 0$$

Since the circle passes through point $(-1, 5)$, then

$$1 + 25 + 10 + 20 + c = 0 \Rightarrow c = -56$$

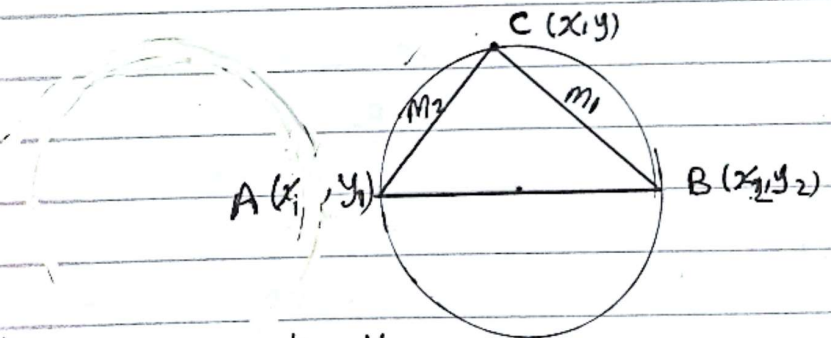
$\Rightarrow$ the equation is

$$x^2 + y^2 - 10x + 4y - 56 = 0.$$

(*) The equation of the circle whose end points of the diameter at $A(x_1, y_1)$, $B(x_2, y_2)$ is given by

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Proof:



$$m_1 = \frac{y - y_2}{x - x_2}, m_2 = \frac{y - y_1}{x - x_1} \quad m_1 m_2 = -1$$

$$m_1 m_2 = \frac{(y - y_2)(y - y_1)}{(x - x_2)(x - x_1)} = -1 \Rightarrow$$

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0.$$

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٢ م

Ex.

Find the center and the radius of the circle which has for a diameter the segment joining the points $(5, -1)$ and $(-3, 7)$

Solution

* equation of the circle

$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0 \Rightarrow$$

$$(x-5)(x+3) + (y+1)(y-7) = 0$$

$$\Rightarrow x^2 + y^2 - 2x - 6y - 22 = 0$$

$$\text{Center } (-f, -g) = (1, 3)$$

$$r = \sqrt{1^2 + 3^2 + 22} = \sqrt{32} = 4\sqrt{2}$$

Ex.

Find the equation of the circle which passing through the three points $(-2, 5)$, $(4, 9)$ and $(2, -1)$

Solution:

The general form of the circle is

$$x^2 + y^2 + 2fx + 2gy + c = 0$$

$\Rightarrow$

$$\text{at } (-2, 5) \Rightarrow -4f + 10g + c = -29$$

$$\text{at } (4, 9) \Rightarrow 8f + 18g + c = -97$$

$$\text{at } (2, -1) \Rightarrow 4f - 2g + c = -5$$

Hence by solving the previous system, we get

$$f = -3 \quad g = -4 \quad c = -1$$

$\Rightarrow$ equation

$$x^2 + y^2 - 6x - 8y - 1 = 0$$

Galaxy M12

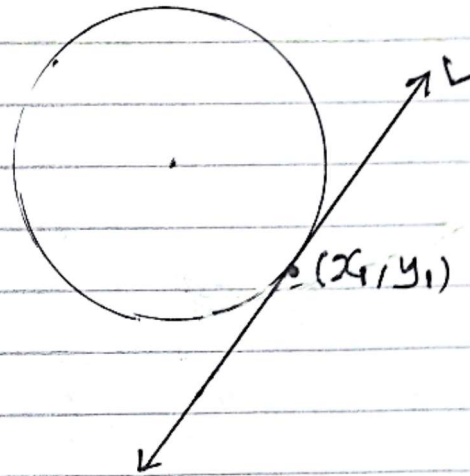
٢٩ يوليو ٢٠٢٥ ١٢:٢٢ م

-6-

No:

Date:

Tangent equation of the circle



Given : (1) eq of the circle
 $x^2 + y^2 + 2fx + 2gy + c = 0 \rightarrow (*)$

(2) tangent point of the straight line L
 is (x_1, y_1)

R.T.P : equation of L

Note that slope of L = $\left(\frac{dy}{dx}\right)_{(x_1, y_1)}$

Now.

differentiating (*) w.r.t x , we obtain

$$2x + 2y y' + 2f + 2g y' = 0$$

$$\Rightarrow y'(y+g) = -(x+f) \Rightarrow y' = -\left(\frac{x+f}{y+g}\right)$$

$\Rightarrow$ If m is the slope of L

$$\Rightarrow m = y'_{(x_1, y_1)} = -\left(\frac{f+x_1}{g+y_1}\right) \rightarrow (1)$$

$$\text{but } m = \frac{y-y_1}{x-x_1} \rightarrow (2)$$

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٣ م

then

$$\text{or } \frac{y-y_1}{x-x_1} = - \frac{(f+x)}{(g+y_1)}$$

$$(x-x_1)(f+x) + (y-y_1)(g+y_1) = 0$$

$\Rightarrow$

$$x x_1 + y y_1 + f(x-x_1) + g(y-y_1) - x_1^2 - y_1^2 = 0 \rightarrow (1)$$

put (x_1, y_1) lies on the circle $\Rightarrow$

$$x_1^2 + y_1^2 + 2fx_1 + 2gy_1 + c = 0$$

$\Rightarrow$

$$-(x_1^2 + y_1^2) = 2fx_1 + 2gy_1 + c \rightarrow (2)$$

from (2) into (1)

$$2x x_1 + 2y y_1 + f(x+x_1) + g(y+y_1) + c = 0$$

which is

the equation of the tangent at a point (x_1, y_1) .

EX.

Find the equation of tangent of the Circle

$$x^2 + y^2 + 8x - 10y + 21 = 0 \rightarrow (*)$$

at the point $(-2, 1)$ and also the perpendicular equation at $(-2, 1)$

Solution

Differentiate (*) w.r.t. x

$\Rightarrow$

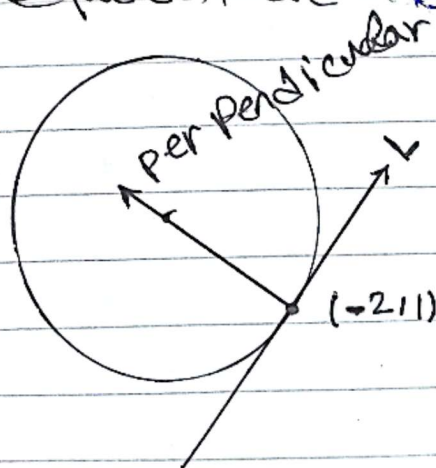
$$2x + 2y y' + 8 - 10y' = 0$$

$\Rightarrow$

$$y' = - \left(\frac{4+x}{y-5} \right)$$

Galaxy M12

$$\Rightarrow m = y'_{(-2,1)} = - \left(\frac{2}{-4} \right) = \frac{1}{2}$$



The equation of the tangent is

$$\frac{1}{2} = \frac{y-1}{x+2} \Rightarrow x-2y+4=0$$

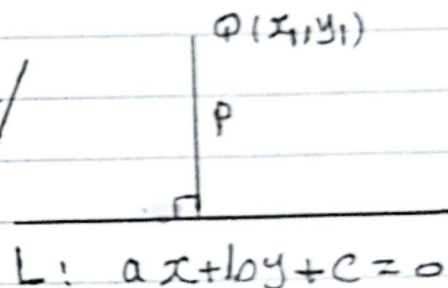
and the perpendicular is

$$\frac{y-1}{x+2} = -2 \Rightarrow 2x+y+3=0$$

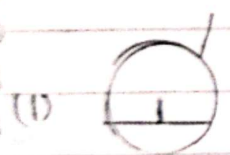
Circle and Straight Line

Def: If P is the distance of perpendicular from the point $Q(x_1, y_1)$ to $L: ax+by+c=0$ then

$$P = \left| \frac{ax_1+by_1+c}{\sqrt{a^2+b^2}} \right|$$

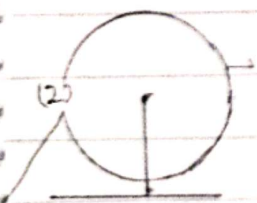


Now: If d is the distance from the center (h, k) of the circle to the line $ax+by+c=0$



$$\Rightarrow d = \left| \frac{ah+bk+c}{\sqrt{a^2+b^2}} \right|, \text{ Let } r \text{ be the radius}$$

1) If $d < r \Rightarrow$ the line intersects the circle at two points



2) If $d > r \Rightarrow$ the line lies out the circle



3) If $d = r \Rightarrow$ the line is a tangent of the circle

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٣ م

-9-

No:

Date:

Ex. Prove that the line $2x+y=11$ intersect the circle $x^2+y^2=26$, and find the two points of intersection.

Solution

$$x^2+y^2=26$$

Center $(0,0)$

$$r = \sqrt{26}$$

$$d: (0,0) \rightarrow 2x+y=11$$

$$d = \left| \frac{2 \cdot 0 + 1 \cdot 0 - 11}{\sqrt{2^2 + 1^2}} \right| = \frac{11}{\sqrt{5}}$$

$d < r \Rightarrow$ the line $2x+y=11$ intersect the circle, To find the intersection points

$$\text{Now } 2x+y=11 \Rightarrow y=11-2x$$

$$\text{From } x^2+y^2=26 \Rightarrow x^2+(11-2x)^2=26$$

$$\Rightarrow (5x-19)(x-5)=0$$

$$\Rightarrow \begin{matrix} x=5 & x=\frac{19}{5} \\ y=1 & y=\frac{17}{5} \end{matrix}$$

Definition: The length of tangent $\overline{PQ}$ to the circle

Given: ① $x^2+y^2+2fx+2gy+C=0$

② $P = P(x_1, y_1)$ lies out the circle

③ $\overline{PQ}$: Tangent at Q

R.T.F $\overline{PQ}$.

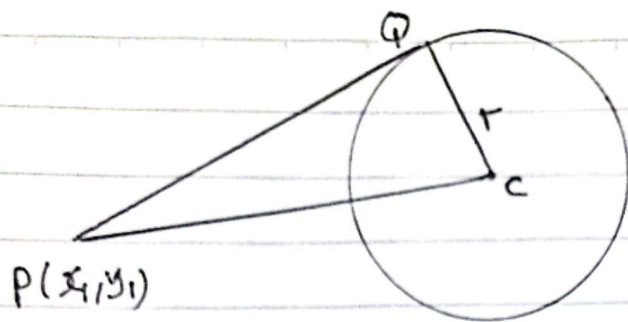
Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٣ م

-/0-

No:

Date:



$$C = (-f, -g)$$

$$\begin{aligned} \overline{PQ}^2 &= \overline{PC}^2 - \overline{CQ}^2 \\ &= (x_1 + f)^2 + (y_1 + g)^2 - r^2 \\ &= x_1^2 + y_1^2 + 2fx_1 + 2gy_1 + f^2 + g^2 - r^2 \\ &= x_1^2 + y_1^2 + 2fx_1 + 2gy_1 + C. \end{aligned}$$

Ex.

Find the length h of tangent to the circle

$$x^2 + y^2 - 4x + 2y - 20 = 0$$

from the point $P(-4, 3)$

Solution

$$(x_1, y_1) = (-4, 3)$$

$$h^2 = x_1^2 + y_1^2 - 4x_1 + 2y_1 - 20$$

$$h^2 = 16 + 9 + 16 + 6 - 20 = 27 \Rightarrow h = 3\sqrt{3}.$$

⑤ Double tangent equation of a circle

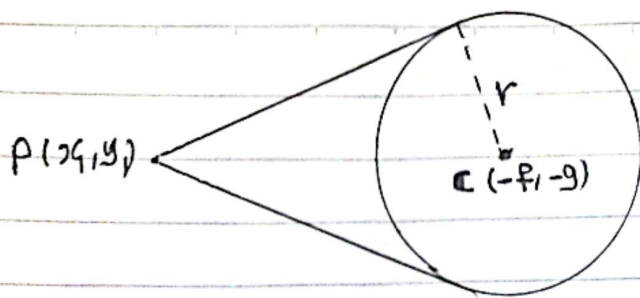
Given: (1) $x^2 + y^2 + 2fx + 2gy + C = 0$

(2) $P(x_1, y_1)$ lies out of a circle

R.T.P. (3) Double Tangent equation of a circle from $P(x_1, y_1)$.

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٣ م



Suppose that the equation of the tangent is

$$y = mx + c \rightarrow (1)$$

$\therefore P(x_1, y_1)$ lies on $y = mx + c$

$$\therefore y_1 = mx_1 + c \rightarrow (2)$$

The distance from center $(-f, -g)$ to the tangent

$$= r = \sqrt{f^2 + g^2 - c}$$

$\Rightarrow$

$$\left| \frac{-mf + g + c}{\sqrt{m^2 + 1}} \right| = \sqrt{f^2 + g^2 - c}$$

$\rightarrow (3)$

$(-f, -g)$	Point
$y = mx + c$	Line
$mx - y - c = 0$	

from (2) and (3) we obtain the values of c and m , and by substituting into (1) we obtain the equation of the tangent.

EX. Find double tangent equation to the circle

$$x^2 + y^2 + 4x - 6y + 8 = 0$$

from the point $(1, 2)$

Solution

$$\text{Since : } x^2 + y^2 + 4x - 6y + 8 = 0 \rightarrow (1)$$

$$\Rightarrow f = 2, g = -3, c = 8, r = \sqrt{f^2 + g^2 - c} = \sqrt{5}$$

$$\text{put } L: y = mx + c \quad (1, 2) \in L$$

$$\Rightarrow 2 = m + c \rightarrow (2)$$

Galaxy M12

٢٩ يوليو ٢٠٢٥ ١٢:٢٣ م

-12-

No:

Date:

$$\begin{aligned} \text{distance from } (-2, +3) \text{ to } L &= r \\ \text{center } mx - y + p &= \sqrt{5} \end{aligned}$$

$$\Rightarrow \left| \frac{-2m - 3 + p}{\sqrt{m^2 + 1}} \right| = \sqrt{5} \rightarrow (3)$$

$$\text{from (2) } p = 2 - m \Rightarrow \frac{-2m - 3 + p}{\sqrt{m^2 + 1}} = \frac{-2m - 3 + 2 - m}{\sqrt{m^2 + 1}} = \frac{-3m - 1}{\sqrt{m^2 + 1}} \rightarrow (4)$$

from (4) into (3) we obtain

$$\left| \frac{-3m - 1}{\sqrt{m^2 + 1}} \right| = \sqrt{5} \Rightarrow \pm(3m + 1) = \sqrt{5(m^2 + 1)}$$

$$\Rightarrow 5(m^2 + 1) = (3m + 1)^2 = 9m^2 + 6m + 1$$

$$= 5m^2 + 5$$

$$\Rightarrow 4m^2 + 6m - 4 = 0$$

$$2m^2 + 3m - 2 = 0$$

$$(m + 2)(2m - 1) = 0 \Rightarrow$$

$$m = -2$$

$$m = \frac{1}{2}$$

$$\text{from } p = 2 - m$$

$$p = 4$$

$$p = 3/2$$

$\Rightarrow$ The equations are

$$y = mx + p \Rightarrow y = -2x + 4 \text{ and } y = \frac{1}{2}x + \frac{3}{2}$$